Full convergence of the speed-preserving Hilbert gradient flow for generalized integral Menger curvature

Candidate solution packet for arXiv:2103.10408

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Status. Candidate full solution, likely valid pending expert review. The theorem below answers the projected-flow convergence question on page 5 of the source paper. It does not address global existence for the ordinary, unprojected ambient Sobolev gradient flow.

Abstract

Let $p \in (7/3, 8/3)$ and $s = 3p/2 - 2 \in (3/2, 2)$. Knappmann, Schumacher, Steenebruegge and von der Mosel constructed a global Hilbert gradient flow for $\mathrm{intM}^{(p,2)}$ projected to preserve the pointwise speed. They proved decay of the projected gradient and weak H^s subconvergence, but left full convergence open. We prove that every one of their trajectories converges strongly in H^s to a projected critical knot. The energy is real analytic by a normalized-chord L^2 factorization. At a critical knot the Blatt–Reiter principal Fourier multiplier is an isomorphism modulo translations and the differentiated remainder is compact, giving a Lojasiewicz–Simon inequality. A Palais–Smale argument supplies a strong accumulation point; the gradient inequality then makes the tail have finite length.

1 The source question

The source is J. Knappmann, H. Schumacher, D. Steenebruegge and H. von der Mosel, *A speed preserving Hilbert gradient flow for generalized integral Menger curvature*, arXiv:2103.10408, later published in *Advances in Calculus of Variations* **16** (2023), 597–635 [4]. Immediately before Corollary 1.4, the authors write that it is not clear whether their solution curves converge to a projected critical point as $t \to \infty$.

For $p \in (7/3, 8/3)$ put

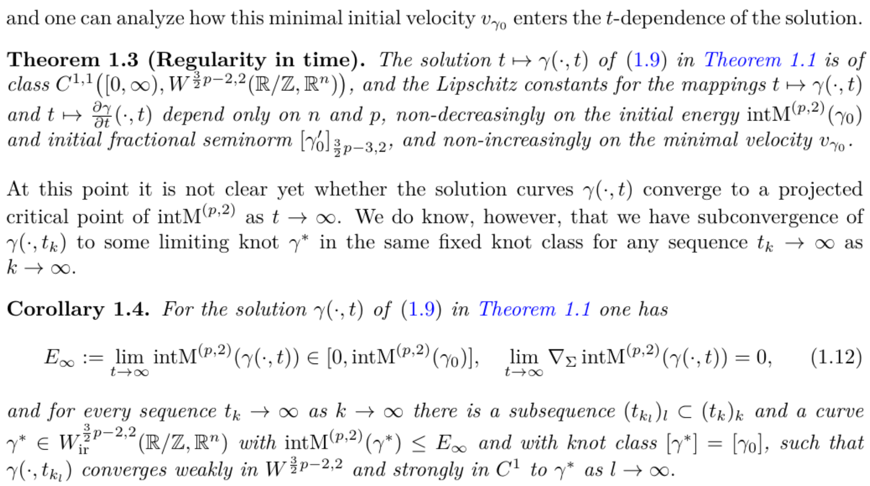
$$s = \frac{3}{2}p - 2 \in \left(\frac{3}{2}, 2\right), \qquad \mathcal{E}(\gamma) = \mathrm{intM}^{(p,2)}(\gamma).$$

The source flow is

$$\dot{\gamma}(t) = -\nabla_\Sigma \mathcal{E}(\gamma(t)), \qquad \Sigma(\gamma) = \log|\gamma'|, \tag{1}$$

where the gradient is taken in H^s and orthogonally projected onto $\ker D\Sigma(\gamma)$. The source proves global existence, preservation of $|\gamma'|$, preservation of the barycenter and knot class, and

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{E}(\gamma(t)) = -\|\nabla_\Sigma \mathcal{E}(\gamma(t))\|_{H^s}^2, \qquad \|\nabla_\Sigma \mathcal{E}(\gamma(t))\|_{H^s} \longrightarrow 0. \tag{2}$$


and one can analyze how this minimal initial velocity v_{γ_0} enters the t-dependence of the solution.

Theorem 1.3 (Regularity in time). *The solution $t \mapsto \gamma(\cdot,t)$ of (1.9) in Theorem 1.1 is of class $C^{1,1}\big([0,\infty), W^{\frac{3}{2}p-2,2}(\mathbb{R}/\mathbb{Z},\mathbb{R}^n)\big)$, and the Lipschitz constants for the mappings $t \mapsto \gamma(\cdot,t)$ and $t \mapsto \frac{\partial\gamma}{\partial t}(\cdot,t)$ depend only on n and p, non-decreasingly on the initial energy $\operatorname{intM}^{(p,2)}(\gamma_0)$ and initial fractional seminorm $[\gamma_0']_{\frac{3}{2}p-3,2}$, and non-increasingly on the minimal velocity v_{γ_0}.*

At this point it is not clear yet whether the solution curves $\gamma(\cdot,t)$ converge to a projected critical point of $\operatorname{intM}^{(p,2)}$ as $t \to \infty$. We do know, however, that we have subconvergence of $\gamma(\cdot,t_k)$ to some limiting knot γ^* in the same fixed knot class for any sequence $t_k \to \infty$ as $k \to \infty$.

Corollary 1.4. *For the solution $\gamma(\cdot,t)$ of (1.9) in Theorem 1.1 one has*

$$E_\infty := \lim_{t\to\infty} \operatorname{intM}^{(p,2)}(\gamma(\cdot,t)) \in [0, \operatorname{intM}^{(p,2)}(\gamma_0)], \quad \lim_{t\to\infty} \nabla_\Sigma \operatorname{intM}^{(p,2)}(\gamma(\cdot,t)) = 0, \tag{1.12}$$

and for every sequence $t_k \to \infty$ as $k \to \infty$ there is a subsequence $(t_{k_l})_l \subset (t_k)_k$ and a curve $\gamma^* \in W^{\frac{3}{2}p-2,2}_{\mathrm{ir}}(\mathbb{R}/\mathbb{Z},\mathbb{R}^n)$ with $\operatorname{intM}^{(p,2)}(\gamma^*) \le E_\infty$ and with knot class $[\gamma^*] = [\gamma_0]$, such that $\gamma(\cdot,t_{k_l})$ converges weakly in $W^{\frac{3}{2}p-2,2}$ and strongly in C^1 to γ^* as $l \to \infty$.

Figure 1: Source PDF page 5. The open convergence statement and the precise weak-subconvergence result are both visible.

2 Main theorem and proof intuition

Theorem 1 (Full convergence)**.** *Let $p \in (7/3, 8/3)$ and let $\gamma_0 \in H^s(\mathbb{T};\mathbb{R}^n)$, $s = 3p/2 - 2$, be a regular embedded closed curve. The global solution of* (1) *constructed in [4] converges strongly in H^s as $t \to \infty$ to a curve γ_∞. Moreover,*

$$\nabla_\Sigma \mathcal{E}(\gamma_\infty) = 0, \quad |\gamma_\infty'| = |\gamma_0'|, \quad \overline{\gamma_\infty} = \overline{\gamma_0}, \quad [\gamma_\infty] = [\gamma_0].$$

Thus γ_∞ is a projected critical knot in the same fixed-speed level and knot class as the initial curve.

Proof intuition

The flow already lives on the correct constraint manifold: its projection is the intrinsic gradient of $\mathcal{E}$ on the level set where speed and barycenter are fixed. The source gives a global bounded trajectory whose gradient tends to zero, but only weak accumulation. The Blatt–Reiter decomposition of $D\mathcal{E}$ has a coercive Fourier-multiplier part of order $2s$ and a genuinely lower-order remainder. It first upgrades every Palais–Smale sequence to a strongly convergent one. At the resulting smooth critical accumulation point, the same decomposition says that the Hessian is an elliptic isomorphism plus a compact operator. Since the energy itself is analytic, the abstract Lojasiewicz–Simon theorem applies. Its gradient inequality bounds the length of the remaining trajectory by a power of the remaining energy drop, so the trajectory can no longer leave a neighborhood of the accumulation point and must converge to it.

3 The preserved constraint manifold

Write $v = |\gamma_0'| > 0$ and $b = \fint_{\mathbb{T}} \gamma_0$. Let

$$\mathcal{M}_{v,b} = \left\{ \gamma \in H^s_{\mathrm{ir}}(\mathbb{T};\mathbb{R}^n) : |\gamma'| = v,\ \fint_{\mathbb{T}} \gamma = b \right\}.$$

The logarithmic-strain results in [4, Section 5] show that this is a smooth Hilbert submanifold. Its tangent space is

$$T_\gamma \mathcal{M}_{v,b} = \left\{ h \in H^s : \langle \gamma', h' \rangle = 0,\ \int_{\mathbb{T}} h = 0 \right\}.$$

Translation invariance gives mean zero for both the ambient and projected gradients [4, Lemma 3.2]. Consequently the right side of (1) is exactly $-\operatorname{grad}_{\mathcal{M}_{v,b}} \mathcal{E}$. Thus projected critical points are precisely critical points of $\mathcal{E}|_{\mathcal{M}_{v,b}}$.

There is no loss in proving the analytic and Fredholm assertions for constant speed. Indeed, with $\ell = \int_{\mathbb{T}} v$ and

$$\phi(u) = \ell^{-1} \int_0^u v(r)\,\mathrm{d}r,$$

composition with the one fixed Sobolev diffeomorphism ϕ maps the constant-speed level onto the v-speed level. Because $s > 3/2$, composition by ϕ and ϕ^{-1} is a bounded isomorphism of H^s. Pulling back the ambient inner product therefore gives an equivalent, time-independent Hilbert inner product. Analyticity, compactness, Fredholm index, Palais–Smale compactness and strong convergence are invariant under this fixed linear change of coordinates. Below we write the constant-speed formulas; the same conclusions hold on $\mathcal{M}_{v,b}$.

4 Analyticity of the energy

The next lemma is the first energy-specific ingredient not recorded in the source literature.

Lemma 2 (Analyticity)**.** *The map*

$$\mathcal{E} : H^s_{\mathrm{ir}}(\mathbb{T}; \mathbb{R}^n) \longrightarrow \mathbb{R}$$

is real analytic.

Proof. We give the factorization because it also explains why the special Hilbert case $q = 2$ is decisive. Let $D \subset (-1/2, 0) \times (0, 1/2)$ be the standard Blatt–Reiter integration triangle. For $a \neq 0$ set

$$A_a \gamma(u) = \frac{\gamma(u+a) - \gamma(u)}{a},$$

with the continuous extension $A_0 \gamma = \gamma'$. Put

$$\mathrm{d}\mu(u, v, w) = |v|^{2-p} |w|^{2-p} |v - w|^{-p}\,\mathrm{d}u\,\mathrm{d}v\,\mathrm{d}w.$$

The prototype fractional estimate in [1, Lemma 4.4] says precisely that

$$B(\gamma)(u, v, w) = A_w \gamma(u) \wedge A_v \gamma(u)$$

is a continuous quadratic map from H^s to $L^2(\mathbb{T} \times D, \mu; \bigwedge^2 \mathbb{R}^n)$. This also follows directly from $A_w\gamma \wedge A_v\gamma = (A_w\gamma - A_v\gamma) \wedge A_v\gamma$ and the Gagliardo characterization of H^{s-1}, using $2s = 3p - 4$.

For an embedded regular curve, every normalized chord is uniformly separated from zero. On a sufficiently small H^s neighborhood of a fixed γ_* there is therefore a $c_* > 0$ such that

$$|A_a \gamma(u)| \geq c_* \quad (u \in \mathbb{T},\ |a| \leq 1/2).$$

Define the bounded multiplier

$$C(\gamma)(u, v, w) = |A_w \gamma(u)|^{-p/2} |A_v \gamma(u)|^{-p/2} |A_{v-w} \gamma(u+w)|^{-p/2} \\ \times |\gamma'(u) \gamma'(u+v) \gamma'(u+w)|^{1/2}.$$

The maps $\gamma \mapsto A_a\gamma$ and $\gamma \mapsto \gamma'$ are linear. Since H^{s-1} is a Banach algebra and embeds into C^0, the analytic Nemytskii maps $z \mapsto |z|^{-p/2}$ away from zero and $z \mapsto |z|^{1/2}$ away from zero show that

$$C : H^s_{\mathrm{ir}} \longrightarrow L^\infty(\mathbb{T} \times D)$$

is analytic locally at γ_*. Multiplication $L^\infty \times L^2(\mu) \to L^2(\mu)$ is continuous bilinear. After the change of variables used in [1, Section 4], the exact energy identity is

$$\mathcal{E}(\gamma) = 6\,\|C(\gamma)B(\gamma)\|^2_{L^2(\mu)}.$$

The right side is a composition of analytic maps. Since γ_* was arbitrary, $\mathcal{E}$ is analytic on all of H^s_{ir}. □

5 Ellipticity and the constrained Hessian

For a constant-speed curve, Blatt and Reiter split the first variation as

$$D\mathcal{E}(\gamma)h = 12Q(\gamma, h) + 12R(\gamma, h). \tag{3}$$

Here Q is independent of the base curve and

$$Q(f, h) = \sum_{k\in\mathbb{Z}} \rho_k \langle \widehat{f}_k, \widehat{h}_k\rangle, \qquad \rho_0 = 0, \qquad \rho_k = c_p|k|^{3p-4} + o(|k|^{3p-4}), \tag{4}$$

with $c_p > 0$ [1, Proposition 4.1]. As $3p - 4 = 2s$, there are constants $c, C > 0$ with

$$c\|h\|^2_{H^s} \le Q(h, h) \le C\|h\|^2_{H^s} \tag{5}$$

on every closed subspace transverse to the constants, in particular on the mean-zero tangent spaces considered here.

Lemma 3 (Compact differentiated remainder)**.** *Let γ be a smooth embedded constant-speed curve. There is a $\delta > 0$ such that the bilinear form*

$$(k, h) \longmapsto D_\gamma R(\gamma, h)[k]$$

extends continuously to $H^s \times H^{s-\delta}$. Consequently it induces a compact operator $H^s \to H^{-s}$.

Proof. We spell out where the gain occurs. Lemma 4.2 of [1] writes R as a finite sum of integrals

$$\int_{\mathbb{T}\times D\times[0,1]^N} G\Big(\frac{|\Delta_w\gamma|}{|w|}, \frac{|\Delta_v\gamma|}{|v|}, \frac{|\Delta_{v-w}\gamma|}{|v-w|}\Big)\,\Gamma_\gamma \otimes h'(u + av + bw), \tag{6}$$

where G is analytic on $(0, \infty)^3$, the remaining factors are shifted tangents, and Γ_γ contains at least two tangent differences with the singular weight $|v|^{2-p}|w|^{2-p}|v - w|^{-p}$. The normalized chords remain in a compact subset of $(0, \infty)$.

Differentiate (6) in direction k. Each summand either (i) differentiates a bounded analytic chord multiplier, (ii) replaces one smooth shifted tangent by k', or (iii) replaces one of the two tangent differences in Γ_γ by the corresponding difference of k'. The fractional product estimate used in the proof of [1, Proposition 4.3], applied with one smooth factor, gives for every

$$0 < \delta < \min\{s - 3/2,\, 2 - s\}$$

the uniform bound

$$|D_\gamma R(\gamma,h)[k]| \le C_{\gamma,\delta}\|k\|_{H^s}\|h\|_{H^{s-\delta}}. \tag{7}$$

For completeness, the power count is the same as in the principal estimate: the two tangent differences absorb the weight of order $2s-2$, while the smooth coefficient permits δ derivatives to be shifted away from the test factor. Cases (i)–(iii) have the same bound; no differentiated term loses both tangent differences. This is also the one-variation version of the differentiated-remainder estimates in [5, Sections 3–4].

Thus the induced operator takes H^s boundedly into $H^{-s+\delta}$. The embedding $H^{-s+\delta} \hookrightarrow H^{-s}$ is compact on $\mathbb{T}$, proving the claim. $\square$

Proposition 4 (Fredholm constrained Hessian)**.** *At every critical point γ_* of $\mathcal{E}|_{\mathcal{M}_{v,b}}$, the Riemannian Hessian*

$$\operatorname{Hess}_{\mathcal{M}_{v,b}} \mathcal{E}(\gamma_*) : T_{\gamma_*}\mathcal{M}_{v,b} \longrightarrow T^*_{\gamma_*}\mathcal{M}_{v,b}$$

is Fredholm of index zero.

Proof. Critical points on the pointwise-speed manifold are also critical under a fixed-length constraint. Indeed, reparametrization invariance annihilates all tangential variations, and a length-preserving variation can be corrected by a periodic tangential variation so that its first-order logarithmic strain vanishes. Hence for some constant λ the curve is critical for $\mathcal{E}+\lambda\mathcal{L}$ on the ambient space. The regularity theorem of Blatt–Reiter makes it smooth [1, Theorem 4].

The constrained Hessian is the restriction of $D^2(\mathcal{E}+\lambda\mathcal{L})(\gamma_*)$ to the tangent space. By (3) it is $12Q$ plus the operator in Lemma 3, plus the second variation of length. The latter has order at most two derivatives and is compact relative to the order-$2s$ operator because $s > 3/2$. The mean constraint eliminates constants, so (5) makes Q an isomorphism from the tangent space to its dual. The Hessian is therefore an isomorphism plus a compact operator. It is Fredholm, and symmetry gives index zero. $\square$

Combining Lemma 2, Proposition 4, and the abstract Hilbert-manifold Lojasiewicz–Simon theorem [2, Theorem 1.4] yields the following.

Proposition 5 (Gradient inequality)**.** *If γ_* is critical for $\mathcal{E}|_{\mathcal{M}_{v,b}}$, then there exist $Z>0$, $\varepsilon>0$ and $\theta\in[1/2,1)$ such that*

$$\|\operatorname{grad}_{\mathcal{M}_{v,b}} \mathcal{E}(\eta)\|_{H^s} \ge Z|\mathcal{E}(\eta)-\mathcal{E}(\gamma_*)|^\theta$$

whenever $\eta\in\mathcal{M}_{v,b}$ and $\|\eta-\gamma_*\|_{H^s}<\varepsilon$.

6 Palais–Smale compactness

Proposition 6. *The restriction $\mathcal{E}|_{\mathcal{M}_{v,b}}$ satisfies the Palais–Smale condition. Equivalently, if $(\gamma_j)\subset\mathcal{M}_{v,b}$ has bounded energy and*

$$\|\operatorname{grad}_{\mathcal{M}_{v,b}} \mathcal{E}(\gamma_j)\|_{H^s} \to 0,$$

then a subsequence converges strongly in H^s to a critical point.

Proof. We again work after the fixed-speed pullback. The energy bound gives a uniform bilipschitz lower bound and a uniform H^s bound by [1, Theorem 1 and Proposition 2.1]. The barycenter constraint controls the constant Fourier mode. Passing to a subsequence,

$$\gamma_j \rightharpoonup \gamma_\infty \quad \text{in } H^s, \qquad \gamma_j \to \gamma_\infty \quad \text{in } C^1.$$

The bilipschitz estimate, speed and barycenter pass to the limit, so $\gamma_\infty \in \mathcal{M}_{v,b}$.

Set $h_j = \gamma_j - \gamma_\infty$. Let P_j be the uniformly bounded projection onto $T_{\gamma_j}\mathcal{M}_{v,b}$ furnished by the logarithmic-strain right inverse in [4, Section 5]. Since the constrained gradients tend to zero,

$$D\mathcal{E}(\gamma_j)(P_j h_j) = o(1).$$

Fix $r \in (3/2, s)$. The same structural estimate as (6), together with the explicit strain right inverse, gives on this energy-controlled set

$$|D\mathcal{E}(\gamma_j)((I - P_j)h_j)| \le C_r \|h_j\|_{H^r} = o(1). \tag{8}$$

This is the Menger counterpart of the projection-defect estimate used in the Palais–Smale proof for tangent-point energy in [3, Section 4]; it follows term by term because $D\mathcal{E}$ contains only differences and derivatives of its test function and the right inverses are uniformly bounded under the energy and speed bounds. The final equality in (8) uses the compact embedding $H^s \hookrightarrow H^r$ together with weak H^s convergence. Hence $D\mathcal{E}(\gamma_j)h_j = o(1)$.

Using (3), bilinearity of Q, and weak convergence,

$$Q(\gamma_j, h_j) = Q(h_j, h_j) + Q(\gamma_\infty, h_j) = Q(h_j, h_j) + o(1).$$

The remainder bound in [1, Lemma 4.2 and Proposition 4.3] gives $|R(\gamma_j, h_j)| \le C\|h_j\|_{C^1} = o(1)$. Therefore $Q(h_j, h_j) \to 0$. The barycenter constraint removes constants, so (5) implies $\|h_j\|_{H^s} \to 0$.

Finally, continuity of the constrained gradient shows that the strong limit is critical. □

7 Proof of full convergence

Proof of Theorem 1. The source estimates make the trajectory bounded in H^s, and (2) says its constrained gradient tends to zero. Choose any $t_j \to \infty$. Proposition 6 provides a subsequence, still denoted t_j, and a critical $\gamma_\infty \in \mathcal{M}_{v,b}$ such that

$$\gamma(t_j) \longrightarrow \gamma_\infty \quad \text{strongly in } H^s.$$

In particular $\mathcal{E}(\gamma_\infty) = E_\infty := \lim_{t\to\infty} \mathcal{E}(\gamma(t))$.

Apply Proposition 5 near γ_∞. Put $e(t) = \mathcal{E}(\gamma(t)) - E_\infty \ge 0$. Whenever the trajectory is in the Lojasiewicz–Simon neighborhood and $e(t) > 0$,

$$\begin{aligned} -\frac{\mathrm{d}}{\mathrm{d}t} e(t)^{1-\theta} &= (1-\theta)e(t)^{-\theta} \|\operatorname{grad}_{\mathcal{M}_{v,b}} \mathcal{E}(\gamma(t))\|_{H^s}^2 \\ &\ge (1-\theta)Z\|\operatorname{grad}_{\mathcal{M}_{v,b}} \mathcal{E}(\gamma(t))\|_{H^s} = (1-\theta)Z\|\dot\gamma(t)\|_{H^s}. \end{aligned}$$

Choose t_j so large that $\gamma(t_j)$ lies in the half-sized neighborhood and $e(t_j)^{1-\theta}$ is smaller than the distance to its boundary times $(1-\theta)Z$. Integrating the displayed inequality shows that the trajectory cannot reach the boundary. Thus it remains in the neighborhood forever and

$$\int_{t_j}^{\infty} \|\dot\gamma(t)\|_{H^s}\,\mathrm{d}t \le \frac{e(t_j)^{1-\theta}}{(1-\theta)Z} < \infty.$$

The tail is Cauchy in H^s and hence converges strongly. Its already known subsequential limit identifies the limit as γ_∞.

The speed and barycenter identities pass to the strong limit. The source paper proves that every finite-time curve is in the initial knot class and gives a uniform energy-dependent bilipschitz bound; strong C^1 convergence therefore keeps the limiting curve embedded and in the same ambient isotopy class. This proves all claims. □

8 Scope and novelty

The theorem concerns exactly the projected flow (1). It does not show that the ordinary ambient flow $\dot{\gamma} = -\nabla\mathcal{E}(\gamma)$ exists for all time. The tempting idea that the two flows differ only by a reparametrization is false at the formal level: the normal correction belongs to $\operatorname{ran} D\Sigma(\gamma)^*$ for the nonlocal H^s inner product and need not be a pointwise tangential vector field.

A bounded literature search through 13 August 2026 used the exact source title and combinations of "full convergence", "integral Menger curvature", "Lojasiewicz–Simon", and "projected critical point". The closest later work is Freches–Schumacher–Steenebruegge–von der Mosel [3], which develops Palais–Smale and strong-subconvergence machinery, and Doehrer–Freches [2], which proves full convergence for *tangent-point* energy. No full-convergence theorem for the projected generalized-integral-Menger flow was found. Novelty confidence is moderate pending specialist review.

References

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